Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

VII Semester B.Tech Biomedical Engineering
End Semester Examinations - November 2024

ELECTIVE - ARTIFICIAL NEURAL NETWORKS [BME4059]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Draw diagrams wherever necessary

1A) Compare an artificial neuron with biological neuron for similarities and differences. (4)

1B) Compare binary and bipolar sigmoidal activation functions. (4)

1C) Construct an artificial neuron analogous to a biological neuron. (2)

2A) Develop a 2-input X-OR function using McCulloch Pitts neuron model. (3)

2B) Explain the major elements of supervised learning network. How its training is different from unsupervised training. (3)

2C) Making use of Euclidian distance and K-means clustering, classify the data {2, 4, 10, 12, 3, 20, 30, 11, 25} into two clusters. Assume 4 and 11 as initial centroids of the clusters. (4)

3A) Train a single output class perceptron with initial weight W^1 for the inputs X_1, X_2 and X_3 with the desired outputs $d_1 = -1$, $d_2 = -1$ and $d_3 = 1$ respectively. Assume a learning rate of 0.1

$$\mathbf{w}^1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0.5 \end{bmatrix} \quad \mathbf{x}_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$ (4)

3B) How do you train an artificial neural network using delta rule? Explain in detail. (4)

3C) Store the pattern [0 1 1 0 1] in a 5-node discrete Hopfield network. (2)

4A) Construct Kohonen Self Organizing Feature Map to cluster four input vectors [0 0 1 1], [1 0 0 0], [0 1 1 0] and [0 0 0 1] into two clusters. Assume an initial learning rate of 0.5 (5)

4B) What are the steps you apply in a design cycle of a neural network application? Explain briefly. (3)

4C) Train an Associative memory network using outer product rule to store the patterns [1, 1, 1, 1] and [-1, 1, 1, -1], and test the network with the test vector [1, 1, 1, 1]. (2)

5A) Write any 4 medical applications of artificial neural networks. (2)

5B) The confusion matrix depicting result of a neural network classifier used for classification of 200 normal and abnormal (Calcification & Blood Clots) CT images is as shown below. Draw inference from the confusion matrix and compute (i) Accuracy (ii) Sensitivity (iii) Specificity (iv) Precision and (V) F1 Score. (3)

		Predicted		
		0	1	
Actual	0	31	19	50
	1	9	141	150
		40	160	200

5C) How do you classify infant cry into 4 classes such as Hunger, Pain, Discomfort, and Pathological using neural network? Explain. (5)

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